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Research paper

The development of a cost model for two supply chain network scenarios for decentralized pyrolysis system scenarios to produce bio-oil

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ABSTRACT

Bio-oil, produced through fast pyrolysis, can act as an intermediate, which can be further upgraded to biofuels. Fast pyrolysis is done in either a centralized or a decentralized (mobile pyrolysis system [MPS]) system. In a centralized system, biomass is transported to a plant to produce bio-oil, which is upgraded in the same unit, while in a decentralized system, the mobile plant is moved to the forest to produce bio-oil, which is transported to an upgrading facility. The objective of this study is to develop supply chain models and conduct a techno-economic assessment of bio-oil production in these two systems using the developed models. For both systems, the bio-oil plant capacity is considered to be 2000 dry t d^{-1}. Because the capacity of an MPS ranges from 10 to 100 dry t d^{-1}, several mobile pyrolysis units are required to achieve the target bio-oil produced in the base case capacity. In this study, models were developed for each supply chain network allocation scenario/configuration for MPS site location (the radial scenario and the truncated scenario) and four relocation time scenarios were assumed (yearly, semi-annually, quarterly, and monthly). Then, bio-oil production costs were evaluated and compared those with a centralized plant. For a 2000 dry t d^{-1} biomass plant, bio-oil production costs are 0.241$ L^{-1}, 0.349 $ L^{-1}, and 0.407 $ L^{-1} for the centralized plant, the MPS truncated scenario (100 dry t d^{-1} MPS unit, yearly relocation), and the MPS radial scenario (100 dry t d^{-1} MPS unit, yearly relocation), respectively.

1. Introduction

Greenhouse gas (GHG) emissions and global warming due to fossil fuel use, the sustainability of biomass (considered carbon neutral), and the availability of local biomass are the main reasons for the growing interest in bioenergy and biofuels.

In North America, there is growing interest in biofuel as a transportation fuel replacement [1]. Canada has 347 million hectares of forest, an estimated 9% of the world's forests [2]. In the last several decades, biomass has been used primarily in the pulp and paper industry, but due to the popularity and adoption of electronic media, demand for pulp and paper is shrinking which was using about 57% of biomass energy [3–5].

Governments around the world support the bioenergy sector and are encouraging the use of biomass resources for the production of biofuels and chemicals [6–10]. Fast pyrolysis is a well-investigated commercial technology for the conversion of biomass to liquid fuels in a centralized plant. Fast pyrolysis takes place in a pyrolyzer in the absence of oxygen at 400–600 °C and at atmospheric pressure [11–13]. The outputs from this process are bio-oil, gas, and biochar. Generally, bio-oil yields are from 50 to 75 wt% depending on the feedstock and process parameters [14,15]. Bio-oil is considered an intermediate for transportation fuel and can be further upgraded through hydroprocessing technology to produce a petro-fuel equivalent [16,17].

In general, centralized plants are used for conventional fuels such as coal and natural gas because of economies-of-scale benefits in capital cost at higher capacities. Usually biomass is transported from the nearest forest to a centralized pyrolysis plant to produce biofuel. The economies of scale concept is also true for biomass [18]. Kumar et al. [18] investigated the production costs of electricity from biomass for different plant capacities. They found that electricity costs initially decreased with an increase in capacity as the economy of scale benefits outweigh the increase in the biomass transportation cost and reach a minimum. With further increases in plant capacity, electricity costs increase. This is because of increases in biomass transportation costs, which outweigh the benefits from the economies of scale in capital cost. Another study by Kumar et al. [19] estimated the cost of power generation from the combustion of Canadian mountain pine-infested wood for a range of plant capacities from 50 to 450 MW and observed that power production costs were highest at low capacity and, with increases in plant capacity, the cost was lowest at 300 MW. But with further increases in plant capacity to 450 MW, costs remained the same.

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Nomenclature

MPS	Mobile pyrolysis system
N	No of MPS units
A	Capacity of centralized plant, dry t d^{-1}
A_1	Bio-oil produced in a centralized plant, dry t d^{-1}
B_1	Capacity of an MPS unit, dry t d^{-1}
Z	Bio-oil used by an MPS to produce electricity, t d^{-1}
M	Number of relocations by an MPS in a year
A_s	Area required for an MPS in relocation time, M ha (relocation time)$^{-1}$
T	Lifetime of an MPS in years
R	Radius of the entire area required for the harvesting for T years
A'	Area required for entire lifetime of all MPS units for T years

Kumar et al. found these same results [18]. Other key challenges for a large-capacity centralized pyrolysis plant are feedstock availability and the need for large capital investment. These challenges might be overcome by using mobile pyrolysis system [20].

The concept of the decentralized or MPS is to reduce the raw biomass handling and transportation costs. The characteristics and advantages of an MPS over a centralized plant are portability, mobility, simplicity, and adaptability to regions and weather.

The technical and economic aspects of a mobile pyrolysis system have been studied and are summarized in Table 1 [21–27]. Brown et al. [22] investigated economic feasibility of two capacities of MPS units, 10 and 50 dry t d^{-1} for small diameter conifer trees found in northern New Mexico forests. A series of experiments were performed simultaneously to enhance pyrolysis oil yield in remote locations. The authors concluded that a larger capacity MPS unit is more cost effective, and the cost of bio-oil can be optimized by reducing labor costs and increasing fixed costs such as suppliers for pre-chipped or pre-sized biomass. A 2013 study focused on two distributed mobile conversion facilities (fast pyrolysis and torrefaction) for a biomass plant capacity of 50 dry t d^{-1} converting forest residue to higher energy density material that can be further processed in a biofuel facility and compared the levelized cost of delivery for these pathways [23]. Badger and Fransham [24] conducted a preliminary techno-economic analysis to estimate the capital cost, including installation cost, of a bio-oil handling system for a mobile pyrolysis system and concluded that the capital cost was comparable to a 50 MWe biomass handling system at the power plant. A techno-economic assessment of 100 dry t d^{-1} southern pine wood chip transportable pyrolysis system was studied by Badger et al. [27] using life cycle operating costs provided by Renewable Oil International (ROI) LLC. To study Fischer-Tropsch fuel production in a centralized catalytic synthesis facility, Wright et al. [21] compared two pathways, the traditional centralized biomass gasification with Fischer-Tropsch synthesis and the distributed pyrolysis system to produce bio-oil followed by bio-oil gasification and Fischer-Tropsch synthesis. For distributed pyrolysis, the authors considered three biomass facilities: on-farm pyrolyzer, a small cooperative pyrolyzer, and a large cooperative pyrolyzer. A few researchers have used the geographic information system (GIS) to optimize the pathway and movement of the distributed pyrolysis system to reduce biomass transportation costs based on feedstock availability and subsequently bio-oil cost [25,26].

All of the studies cited above focus on the technical and economic feasibility of a mobile pyrolysis system to produce bio-oil. But there is no study available in the literature describing the development of a supply chain network model for the allocation of MPS units over the lifetime of the operation. The models developed here assess MPS units of five different capacities (10, 20, 40, 50 and 100 dry t d^{-1}) and four relocation times (yearly, semi-annually, quarterly, and monthly) to estimate bio-oil production costs. Relocation time refers to the frequency with which a decentralized unit moves in a year (i.e., semi-annually relocation means that MPS unit will move 2 times in a year). To create a framework for the distributed pyrolysis system, two configurations are considered and are shown in Fig. 1. The results are compared with those from the centralized pyrolysis facility.

The key objectives of this study are:

- The development of supply chain networks to evaluate the distributed mobile system's bio-oil production through pyrolysis process using Canadian hardwood as the feedstock for a base case plant capacity of 2000 dry t d^{-1},
- The development of techno-economic models for each MPS configuration (radial and truncated) to assess their economic feasibility,
- The comparison of bio-oil production costs from both centralized and distributed pyrolysis systems for the base case capacity of 2000 dry t d^{-1},
- The evaluation of the optimum plant capacity for each configuration at the point at which they intersect with the centralized system.

In this study, the terms scenario and configuration are used interchangeably, just as decentralized and mobile pyrolysis system are.

2. Method

2.1. Centralized pyrolysis plant

A centralized pyrolysis plant handles the large-scale production of bio-oil from whole tree biomass feedstock at a single facility (a fixed facility where biomass is brought from nearby), as shown in Fig. 1 (c). A centralized facility is located at the center of a circular area and biomass is transported to the plant gate for processing. The biomass is harvested, chipped, and then trucked to the plant gate. The chipped biomass is then ground to a particle size of less than 2 mm and dried to a moisture content of less than 10 wt%, which is suitable for the pyrolyzer [12]. Biomass pyrolysis takes place in a fluidized bed reactor in the absence of oxygen with nitrogen as the fluidizing gas to produce vapor and biochar. Biochar is separated through a cyclone and combusted to generate energy. The pyrolysis vapor is quenched in a series of condensers to produce bio-oil. For the centralized plant, it was assumed that the pyrolysis unit and upgrading unit are located at the same place, so bio-oil transportation cost was not considered in the analysis. The detailed process and operating conditions of a centralized plant are further explained by Sarkar and Kumar [28].

2.2. Mobile pyrolysis system

As a mobile pyrolysis plant is convenient, easy to assemble and use, and portable, it is considered a strong alternative to a centralized plant. An MPS unit is loaded on a trailer that is attached to a truck or tractor. The unit can be moved to a remote area where feedstock is available and the produced bio-oil can be transported to an upgrading facility. As the density of bio-oil (1.2–1.3 kg L^{-1}) is almost three times greater than the density of raw biomass, transportation costs are low [28]. An MPS unit consists of a dryer, grinder, biomass feeder, auger screw feeder, auger reactor, cyclone separator, condenser, biochar collector, and bio-oil collection chamber (see Fig. 2). Each MPS unit requires a biomass field storage area.

Auger pyrolysis technology was developed by ABRI-Tech [29]. The reactor mechanically mixes the biomass and uses a heat transfer medium. In the pyrolysis system, two independent rotating devices rotate inside the horizontal auger reactor; the reactor does not autorotate [20,22]. The heating medium is heated prior to pyrolysis, so no

Table 1
Research on MPSs.

MPS capacity	Process	Feedstock	Location	Comment/References
10 and 50 dry t d^{-1}	Fast pyrolysis	Small diameter conifer tree	Northern New Mexico	Labour cost and the increased use of fixed MPS unit are the major driving factors to control bio-oil cost. 50 dry t d^{-1} performs better than 10 dry t d^{-1} [22].
50 dry t d^{-1}	Fast pyrolysis and torrefaction	Forest residue	Canada	Forest residues can be converted to a useful form (bio-oil/torrefied wood) through a distributed conversion facility (where they undergo fast pyrolysis or torrefaction) and transported to a biofuel facility. Initial moisture content, transport distance, and the haul cost are the deciding factors for the levelized delivered cost of feedstocks [23].
20000 gallon bio-oil	Fast pyrolysis	Green woodchips	Renewable Oil International, USA	A preliminary techno-economic analysis was conducted to estimate the capital cost, including the installation cost, of a bio-oil handling system for a mobile pyrolysis system and the researchers concluded that the capital cost was comparable to a 50 MWe biomass handling system at a power plant [24].
100 dry t d^{-1}	Fast pyrolysis	Southern pine wood chips	Renewable Oil International, USA	The production cost of bio-oil was reported to be 1.08 $ gal^{-1} (US dollar). Bio-oil production cost is drastically influenced by feedstock cost, feedstock quality. and pretreatment of biomass (ash content) [27].
40 dry t d^{-1}	Fast pyrolysis	Corn stover and energy sorghum	Texas, Illinois and Nebraska USA	For a 40 dry t d^{-1} mobile pyrolysis system, three biomass feedstocks and five relocation frequencies (yearly, semi-annually, quarterly, monthly and stationary) were analyzed. The probability of success increases with reductions in relocation frequency [25].

fluidizing medium is required, thereby further increasing the efficiency of the condensers as they handle only the pyrolysis vapor. Biochar is collected from two places: at the end of the reactor and from the cyclone char collector. In this study, it is assumed that non-condensable gases are the heating carrier for the reactor. The pyrolysis vapor products are separated from the reactor due to the pressure difference and passed through the cyclone and a series of condensers where the bio-oil is collected and non-condensable gases are combusted to produce heat (and used as the heating carrier). As the MPS unit is located in the forest, some bio-oil is used to produce electricity using a flex fuel generator [20].

The main objective of this paper is to develop generic models to investigate the technical and economic feasibility of bio-oil production from two MPS configurations. A base case of a 2000 dry t d^{-1} biomass facility was considered and Canadian hardwood were assumed as feedstock. The capacity of an MPS unit is from 10 to 100 dry t d^{-1}, thus many units are required to achieve the target base case capacity. The actual number of units required is higher than expected because some of the produced bio-oil is used to produce electricity, which affects the overall revenue. For each MPS, a restricted area is assigned for the lifetime of the harvesting operation so that no two MPSs intersect. In this study, five MPS capacities (10, 20, 40, 50, and 100 dry t d^{-1}) and four relocation times (yearly, semi-annually, quarterly, and monthly) were considered. Two scenarios (configurations) were assumed in the allocation of harvesting areas, truncated and radial; they are shown in Fig. 1 (a) and (b), respectively. Two generic equations were developed to estimate bio-oil production costs for the two configurations, and the results were compared with those reported by Sarkar et al. [28] for a centralized pyrolysis system.

2.2.1. Generic model development for an MPS

Fig. 1 (a) and (b) show two MPS scenarios in which all the units operate simultaneously to achieve the target base case plant capacity. To derive generalized equations, each parameter is considered as a variable that can be changed depending on the requirements. It is assumed that a centralized plant of A dry t d^{-1}capacity produces A_1 dry tonnes of bio-oil each day. One MPS unit of B dry t d^{-1} capacity yields B_1 dry tonnes of bio-oil each day. Because an MPS operates in a remote location, some of the produced bio-oil, z t d^{-1} of bio-oil, is used to generate electricity. To ensure equal amounts of bio-oil from each framework (centralized and MPS) for comparison, N numbers of MPS units are required and can be calculated from Eqn. (1):

$$N = \frac{A_1}{B_1 - z} \tag{1}$$

The required biomass area for one MPS in each relocation time is M ha (represent by A_s) and the lifetime of each MPS is T years. It is important to note that the area required for each MPS for the lifetime of the operation is assigned at the beginning of the operation to avoid overlapping between MPS units. In addition to M ha area, each MPS requires ¾ acre (0.3035 ha) for its set-up [25]. Therefore, the total area required by each MPS is $M + 0.3035$ ha. For example, in the yearly relocation scenario, the MPS requires M ha/year, and the total area required for a 10-year MPS lifetime is $10*(M + 0.3035)$ ha. The same area is required by each MPS.

The total area A' required for N MPS units for the lifetime of the operation for both scenarios can be estimated using Eqn. 2:

$$A' = (M + 0.3035)*N*T*m \tag{2}$$

where m is the number of MPS relocations in a year (e.g., $m = 1$, 2, 4, and 12 for yearly, semi-annually, quarterly, and monthly relocations, respectively).

The area required for the lifetime of all MPSs is considered to be circular because for a centralized plant, it is also assumed that the area is circular and the plant is situated at the center of the region. The radius of the circular area R is calculated using Eqn. (3):

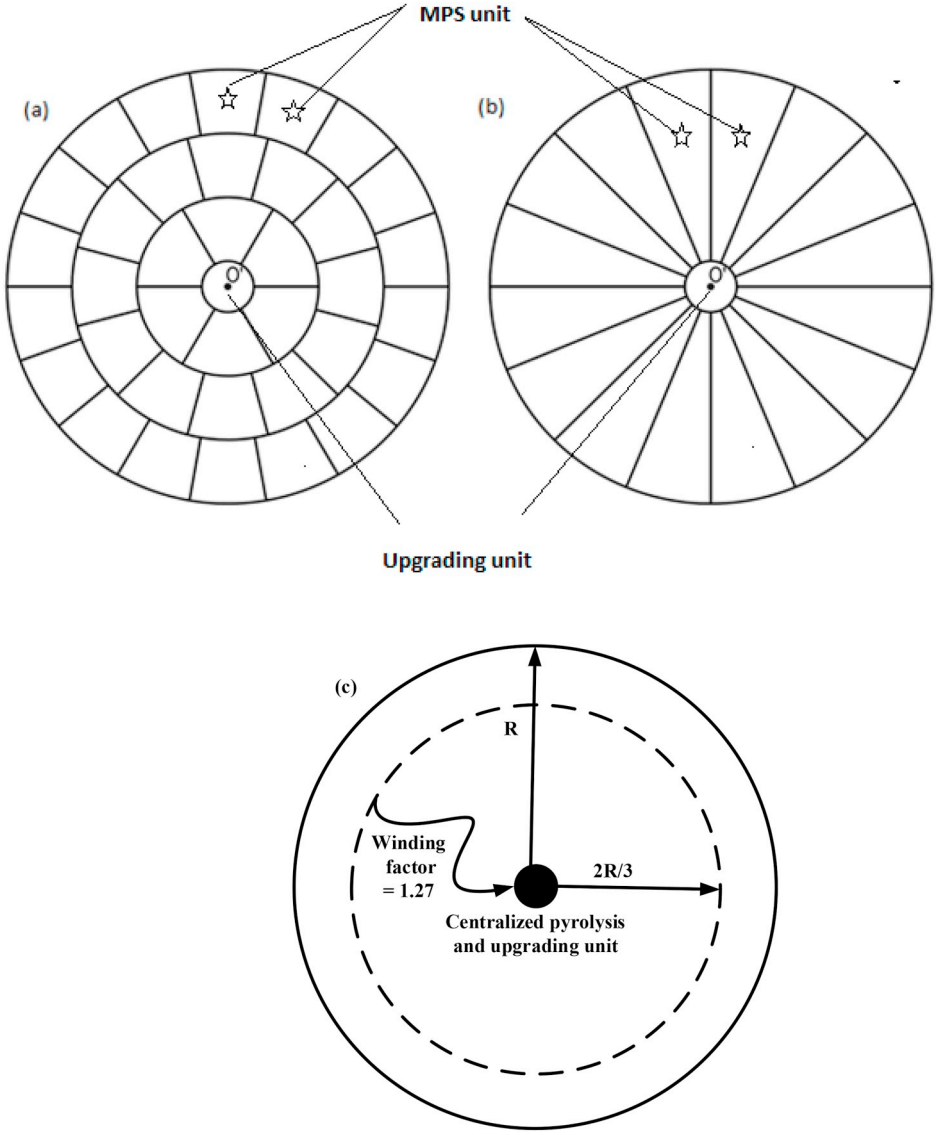


Fig. 1. Two MPS allocation configurations: (a) truncated and (b) radial and (c) centralized unit.

$$R = \left(\frac{2}{3}\right)*1.27*\sqrt{\frac{(M + 0.3035)*N*T*m}{\pi}} \quad (3)$$

R is constant for the two MPS scenarios (truncated and radial), but the calculations for the other parameters, such as biomass transportation distance to the MPS unit, bio-oil transportation distance from the MPS to the bio-oil processing facility, and relocation distance, are explained in the subsequent section. In Eqn. (3), 1.27 is the winding factor (used to account the winding of the roads in a remote area) and 2/3 is the average transportation displacement of the biomass collection area [30].

2.2.1.1. Scenario 1 – truncated. In the truncated scenario, the total circular area of radius R is divided into rings (see Fig. 1 (a)) and each MPS is placed in the truncated d segment. The upgrading unit would be located at the center O' of the circular area of radius R. Therefore, the truncated d segments in the innermost ring do not touch the center (it will still be a truncated area). The section framed by the letters *A, B, F, C, D,* and *E* is the location area, A_s, for one MPS for a particular relocation time in which $AD = BC = EOF = d$, as shown in Fig. 3. In the segment, the MPS unit is located at the center O of the polygon *ABFCDEA*. Eqn. (4) is the derived expression of A_s to calculate d. The detailed calculations steps are given in Appendix A (Fig A1)

$$A_s = \frac{2\pi Rd - \pi d^2}{2\pi} * \frac{d}{\left(R - \frac{d}{2}\right)} = d^2 \quad (4)$$

From Eqn. (4), the distance d can be calculated using Eqn. (5):

$$d = \sqrt{(M + 0.3035)} \quad (5)$$

2.2.1.1.1. Bio-oil transportation distance. The bio-oil transportation distance is the distance bio-oil is transported from the MPS unit to the central bio-oil processing facility at the center O' (see Fig. 4(a)). It is important to estimate the number of relocations (A_s) based on N MPS units and allocate the position of each MPS so that the MPS units do not interact with each other during the lifetime of each plant. Fig. 4(a) shows the formation of rings and the division of each ring location area (A_s) for each MPS unit and '•' shows the position of the MPS units in the A_s area. The innermost circular area is for the upgrading unit to process the bio-oil to produce renewable diesel.

The number of rings (N') can be calculated from Eqn. (6):

$$N' = \frac{R}{d} \quad (6)$$

The number of A_s in the circle with the radius R ($N_{N'}$) can be calculated using Eqn. (7), which is the sum of the all the A_s in all the rings.

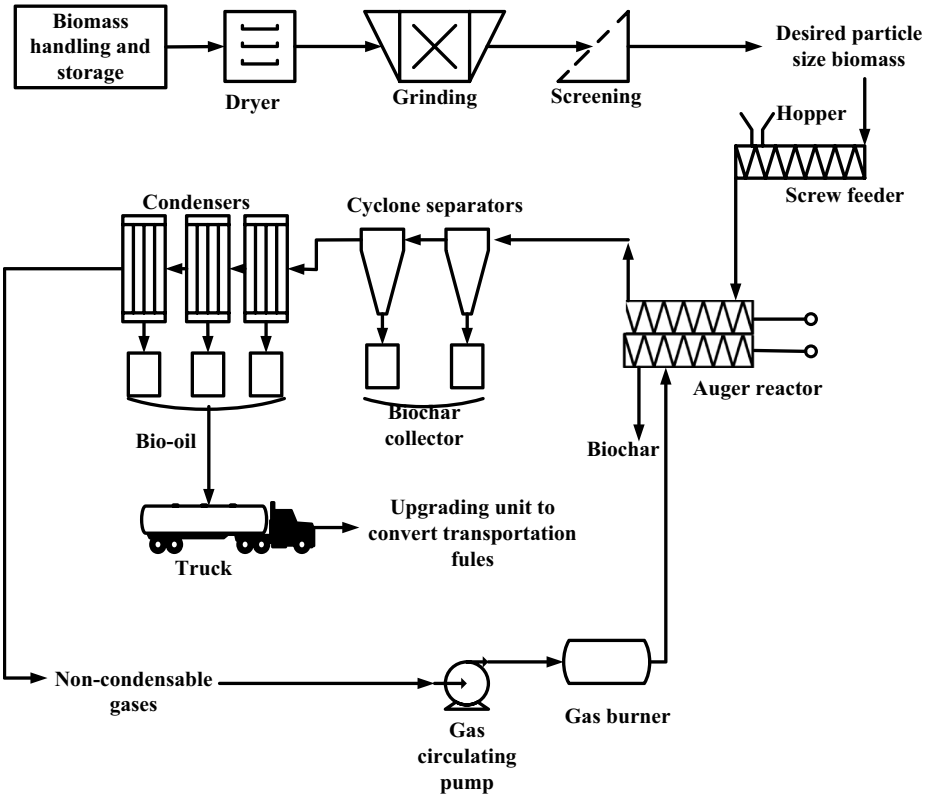


Fig. 2. Schematic of an MPS unit.

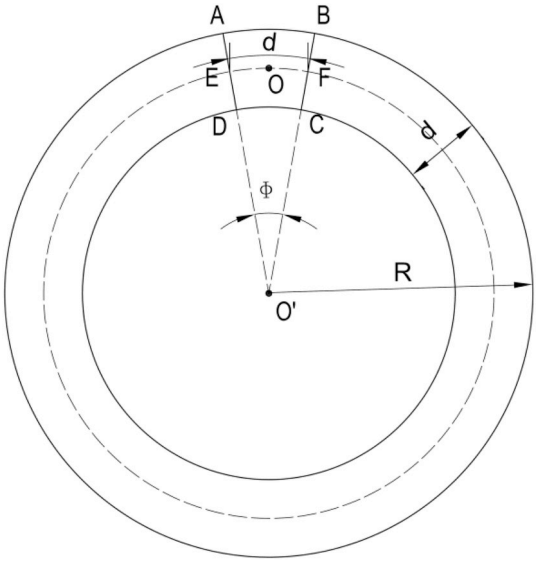


Fig. 3. MPS allocation in area A_s for scenario 1: Truncated.

The detailed calculation is given in Appendix A (Figure A2),

$$N_i' = \sum_{i-1}^{N'} \frac{Perimeter\ of\ dotted\ circle\ N_i'}{d} = \sum_{1}^{N'} \frac{2\pi\left(R - \frac{(2i-1)d}{2}\right)}{d} \quad (7)$$

where i = 1, 2, 3 … N'.

The bio-oil produced in the MPS needs to be transported to a bio-oil processing facility. Bio-oil transportation distances vary with the location of the MPS.

Eqn. (8) is the generalized expression for the bio-oil transportation distance for all A_s located in the N' rings. D_N is the total bio-oil transportation distance for N MPS units for the entire lifetime of the plant for different relocations. The distance covered each year can be calculated by taking 1/10 of the D_i since 10 years is considered the MPS lifetime in this study.

$$D_N = \sum_{i=1}^{N'} N_i'^{*}\left(R - \frac{(2i-1)d}{2}\right) = \sum_{i=1}^{N'} \frac{2\pi\left(R - \frac{(2i-1)d}{2}\right)}{d} \cdot \left(R - \frac{(2i-1)d}{2}\right) = \sum_{i=1}^{N'} \frac{2\pi\left(R - \frac{(2i-1)d}{2}\right)^2}{d} \quad (8)$$

2.2.1.1.2. Biomass transportation distance. Biomass transportation distance varies with relocation time and MPS capacity. In each A_s, the MPS is located at the center (O) and biomass is transported from the surrounding area, as shown in Fig. 4(b). To estimate average biomass transportation distance D for a relocation period, the truncated area *ABFCDEA* is divided into two segments: one is *ABFEA* (outer arc to MPS) and the second is *EFCDE* (MPS to inside arc). To estimate the average biomass transportation distance, a small strip of area dx located at a distance of x from the center $O'$is considered for both segments as shown in Fig. 4(b). The small area dx, known as *PQRST*, is divided into two equal polygons, *PQR* and *RST*. *Q* and *S* are the biomass collection centers following harvesting, skidding, and chipping. One 10 dry tonne capacity truck per trip is used to deliver the feedstock to the MPS from the biomass collection point [31]. Generally for a centralized plant, 20–40 tonnes of biomass are transported per trip from the roadside to the plant gate [32]. But for a decentralized plant, biomass need to be transported from a remote forest location to the MPS unit. Therefore, in this study, a small capacity truck of 10 dry tonne is considered [30–32]. The paths followed are *QR* and *RO* for the small polygon *PQR* and *RS* and *RO* for the other side of the polygon *RST*. The average biomass transportation distance for the truncated area *ABFCDEA* is given in Eqn. (9). The detailed derivation steps are given in Appendix A.(Figure A3)

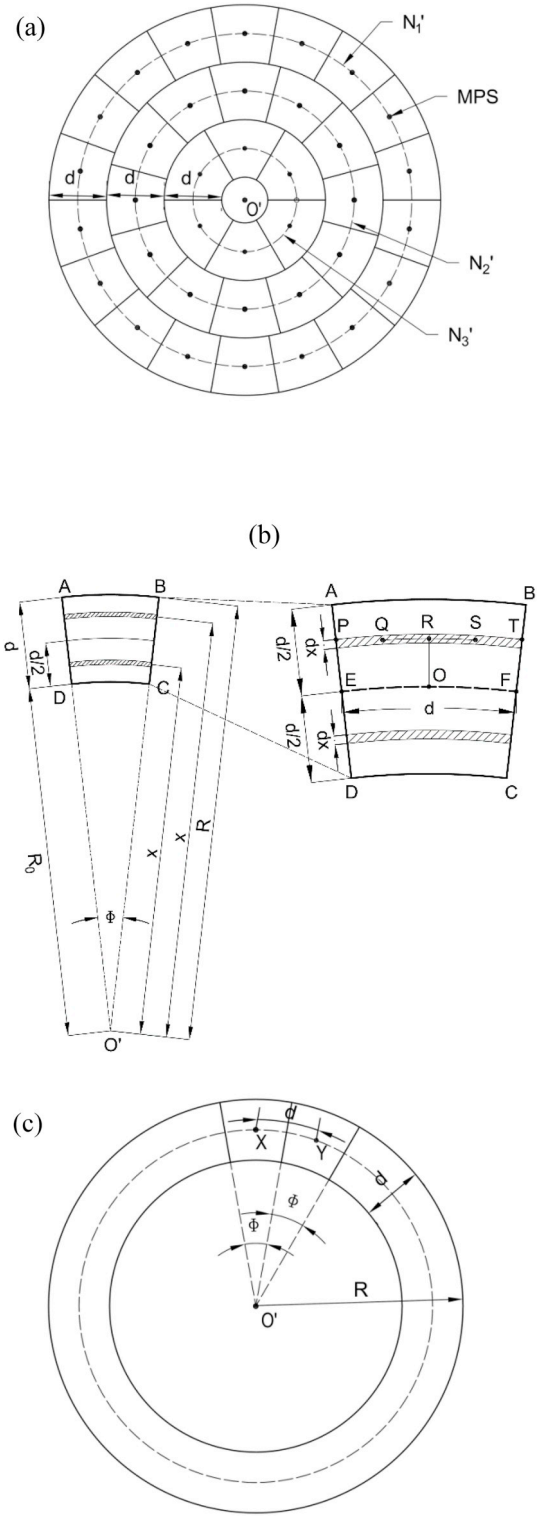


Fig. 4. Bio-oil transportation distance (a), biomass transportation distance (b) and MPS relocation distance (c) for the truncated scenario.

$$D = \frac{7d^3}{10(d + 2R_0)^2}[13d^2 + 40dR_0 + 36R_0^2] + \frac{7d^3}{5(d + 2R_0)}\left[\frac{d}{2} + 3R_0\right] \tag{9}$$

2.2.1.1.3. Relocation distance. Relocation distance refers to the distance travelled by the MPS between two consecutive allocated points where it operates and is shown in Fig. 4(c). *X* and *Y* represent the MPS locations for the respective relocation times. The distance travelled by the MPS from point *X* to point *Y* is *d*.

2.2.1.2. Scenario 2 – radial. Fig. 1 (b) shows the radial configuration for MPS operations. In this scenario, the number of MPSs required and the radius of the circular area are estimated through Eqn. (1) and Eqn. (3), respectively. BO'C is the area required by one MPS throughout its lifetime and is shown in Fig. 5.

Fig. 5 also shows the distribution of an MPS unit's area into subsectors based on relocation frequency. An MPS will relocate with time in the assigned area. In each relocation, the MPS settles in the centroid position (the average distance from every point in a specified area) to minimize biomass and bio-oil transportation distance. In this configuration, each sector is assigned by dividing the total angle (2π) with the number of MPS units and relocation frequencies. The angle β is subtended by each sector and can be calculated by Eqn. (10).

$$\beta = \frac{2\pi}{Nm} \tag{10}$$

2.2.1.2.1. Bio-oil transportation distance. The bio-oil produced from each MPS is transported to the bio-oil upgrading facility, as shown in Fig. 6(a). In sector PO'Q, the MPS is situated at the centroid (*O*) before being moved. Therefore, the bio-oil transportation distance, *O′O*, is two-thirds of the radius of the entire area, as shown in Eqn. (11).

$$O'O = R' = \frac{2R}{3} \tag{11}$$

2.2.1.2.2. Biomass transportation distance. Fig. 6(b) shows the biomass transportation distance in the radial configuration for a subsector at a particular relocation period. For the transportation distance, the sector *PO′Q* is divided into two segments: one is from the center *O′* to the centroid and the second is from the centroid to the end of sector *PO′Q*. To calculate the average biomass transportation distance, a small strip of area d*r* is considered in both segments, shown in Fig. 9. The small area d*r*, known as polygon *WXBNELKMW*, is located at distance *r* from the center. The biomass in the area d*r* is divided into two equal polygons, *WXBNMW* and *MNELKM*. *B* and *E* are the biomass collection centers following harvesting, skidding, and chipping. A 10 dry tonne capacity truck is used to deliver the feedstock to the MPS from the biomass collection points. The total average distance travelled by truck can be estimated by summing the two integrations for the sector PO'Q, which are estimated in Eqn. (12). The deatiled calculation is given in Appendix A.(Figure A4)

$$\mathrm{D} = \frac{2*84*2\pi}{20*Nm} * \frac{R^3}{27}\left[\left(\frac{9*2\pi}{4Nm}\right) + \frac{8}{3}\right] \tag{12}$$

2.2.1.2.3. Relocation distance. Fig. 6(c) shows the two adjacent segments, PO'Q and QO'S, where one MPS unit is moved following the completion of a harvest. The relocation distance from PO'Q to QO'S is the sum of *KC* and *CL* and is represented by Eqn. (13). We assumed that O'CK and O'CL are a right-angled triangle.

$$\textit{MPS Relocation distance} = X = KC + CL = 2r = 2R'sin\left(\frac{\beta}{2}\right) = 2*(2R/3)*\sin\left(\frac{2\pi}{2Nm}\right) \tag{13}$$

2.3. Description of economic parameters

This section describes the techno-economic parameters used in the MPS analysis. All the cost numbers are reported in 2017 US dollars (1 USD = 1.33 CAD). Biomass properties are given in Table 2 [18,33]. The cost model inputs used to estimate bio-oil production costs from the MPS are shown in Tables 3 and 4. The characteristics of the centralized 2000 dry t d^{-1} plant were taken from Sarkar and Kumar [28].

2.3.1. Field cost

The field cost includes harvesting (cutting, skidding, and chipping),

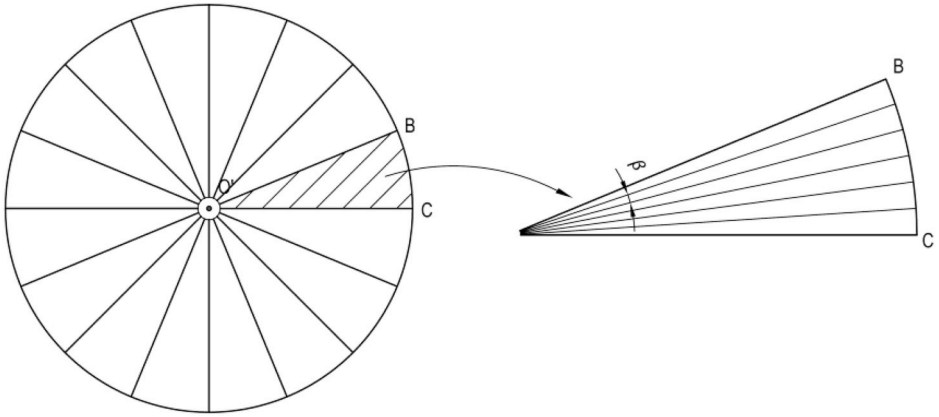


Fig. 5. Distribution areas into subsectors, radial scenario.

the premium paid to the land owner, road construction cost, and silviculture cost [34]. In this study, it was assumed that harvesting equipment (such as a feller bencher for cutting, a skidder to skid whole trees, and a chipper for making chips) is rented and was considered operating cost. A premium of 5.26 \$ t^{-1} of feedstock is paid above the forest biomass cost for the collection and use of the forest biomass [35]. The silviculture cost is the cost to cultivate and grow new trees after harvesting and is 119.04 \$ ha^{-1} [35]. Road constuction costs are the costs to build a tertiary road network that can be used during harvesting and for truck capacities of 10 dry tonnes [35,36]. The details of these costs are listed in Table 3 [18,23,24,31,34,35,37–39].

2.3.2. Biomass and bio-oil transportation cost

Biomass and bio-oil transportation costs comprise two components, the fixed cost for loading and unloading and the variable cost for the round-trip transportation of biomass and bio-oil [34]. The variable cost is a function of plant capacity. In both scenarios, both biomass and bio-oil transportation distance were shown to be a function of the radius of the whole circular area. The radius depends on the yield of the biomass as a higher yield reduces the area required for the plant.

2.3.3. Capital cost

The capital cost is the investment to fabricate an MPS unit to produce bio-oil in a remote location. For this analysis, five capacities are considered: 10, 20, 40, 50, and 100 dry t d^{-1}. Capital cost data were drawn from publications and industrial reports [20,25,40]. From the available data, a scale factor was derived to calculate the capital cost at various capacities. As the MPS concept is new and operation is available only on a small scale in a laboratory setting, there is very limited published research available. The MPS unit characteristics and cost inputs are given in Table 4 [18,20,22,28].

2.3.4. Operational cost

The operational cost includes labor, maintanance, energy, and relocation costs. Each MPS unit operates continuously for 24 h per day. There are 3 eight-hour shifts and each is run by 2 employees (one for controlling and one for fuel handling) for an MPS capacity of 10–50 dry t d^{-1}, and one adminstrative staff is assigned to each MPS. But for 100 dry t d^{-1} MPS unit, one extra employee is provided per shift, which increases the number of employees to 3 (one for controlling and two for fuel handling). An average wage of 30 \$ h^{-1} is assumed; this includes benefits [18,20,28]. The maintanance cost is assumed to be 3% of the initial capital cost investment including all the MPSs (equivalent to a base case capacity of 2000 dry t d^{-1}) [18]. The relocation cost is the cost to move an MPS unit from one point to another by two rental trucks. Relocation involves three activities: dismantling, moving the unit to the next processing point, and reassembling. It is assumed that it takes 6 h to dismantle the unit, 5 h to move it, and 6 h to reassemble the unit, or 17 h altogether [20]. The rental truck charge is considered to be 78 \$ h^{-1}. Electricity from bio-oil, noncondensable gases, and propane are used as the energy in the plant. Thermal energy for pyrolysis is provided by the non-condensable gases and propane in the ratio of 3:1, respectively. The details of the operating costs are in Table 4.

2.3.5. Biomass and bio-oil storage costs

It is assumed that biomass is stored for at least 15 days. On-field storage costs are 1.1 \$ t^{-1} [35]. The bio-oil produced from the MPS unit is stored on site for 3 days in a 100 m^3 capacity stainless steel tank and then transported to the bio-oil processing plant in a 30 m^3 capacity liquid tank [38]. Storage cost details are provided in Table 3.

2.3.6. Biochar transportation cost

The biochar produced during pyrolysis is spread out in the harvested area to preserve soil nutrients for reforestation [41]. Costs to transport the biochar from the plant gate to the field are 4.5 \$ t^{-1} of biochar [37].

2.3.7. Product cost

Although bio-oil is ultimately the source of revenue from the MPS unit, some of it is used to produce electricity for the portable pyrolysis unit. The bio-oil production cost is calculated using a discounted cash flow method.

2.3.8. Uncertainty analysis

In this study, a detailed robust approach has been employed to estimate bio-oil production cost from two distributed pyrolysis facility configurations. But, as we know, the mobile pyrolysis process is an emerging technology and not yet commercialized, and so a degree of uncertainty is associated with the field cost and operating parameters of an MPS unit. To reduce the uncertainty and risk associated with the input parameters, a uncertainty analysis was performed using the Monte Carlo application [42] on the base case plant capacity of 2000 dry t d^{-1}. The idea behind this simulation is to use the randomness of the input variables to obtain an accurate result without propagating errors. Each variable is changed in a range depending on its sensitiveness on the output and large numbers of iterations are used to produce accurate results.

3. Results and discussion

This study develops techno-economic models to investigate the economic and technical feasibility of replacing centralized pyrolysis plants with a decentralized system. A centralized plant benefits from economies of scale, but mobile pyrolysis, because it can reduce biomass transportation costs and improve accessibility in remote locations and in extreme weather conditions, is gaining some interest. For the base

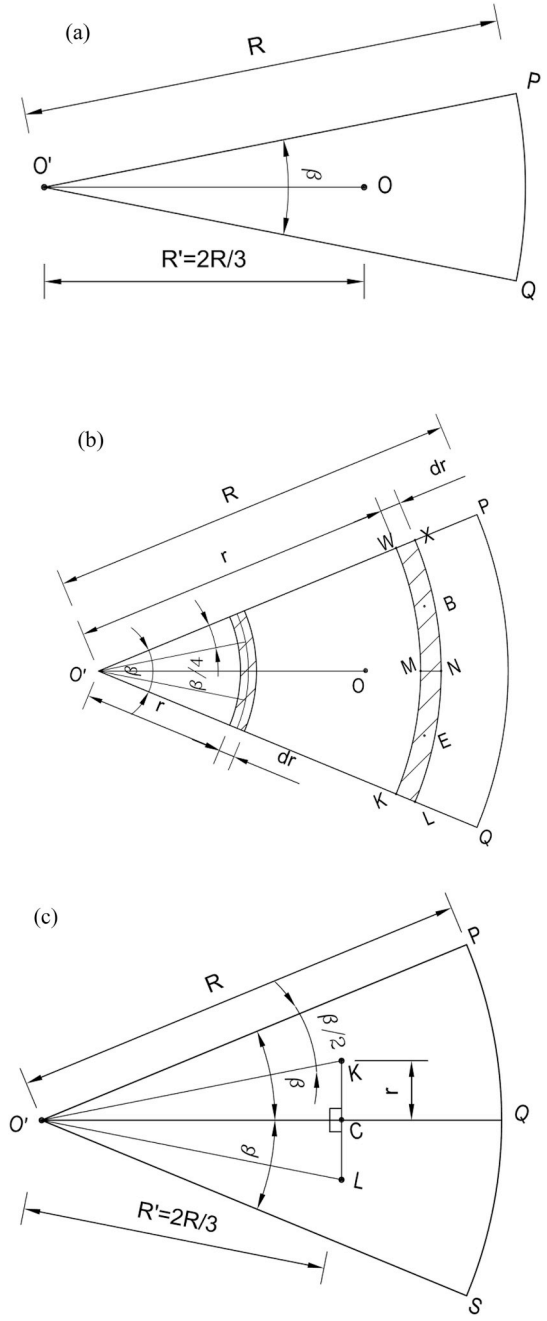


Fig. 6. Bio-oil transportation distance (a), biomass transportation distance (b), and MPS relocation distance (c) for the radial scenario.

Table 2
Whole tree biomass properties.

Parameters	Value	Reference/Comments
Feedstock type	Hard wood	This study
Moisture content (%)	50	[18]
Heating value (MJ kg^{-1}, HHV)	20	[18]
Fuel density (kg m^{-3})	486	[33]
Biomass yield (t ha^{-1})	84	[18]

case comparison, a centralized plant with a capacity of 2000 dry t d^{-1} producing 72% by weight of bio-oil was assumed [28]. To produce a comparable amount of bio-oil per day from an MPS, several units operating simultaneously are required.

3.1. Estimate of the number of MPS units required for different capacities

Fig. 7 shows the number of MPS units required with changes in plant capacity. Both scenarios require the same number regardless of plant capacity. A significant part of the bio-oil produced is used to generate electricity for biomass pretreatment in the MPS. 31.25 gallons of bio-oil produces 0.75 MM BTU of electricity per hour [20]. This reduces the bio-oil revenue from 57% to 50% by weight for each MPS. Therefore the actual number of MPS units required for the various capacities is not an exact divisible relation to the base capacity of 2000 dry t d^{-1}. When the MPS capacity increases, the required number of MPS units decreases.

3.2. MPS capital costs

Few studies have been done on the mobile pyrolysis process. To develop a scale factor to calculate the capital cost of an MPS unit, a curve was plotted showing the capital cost and MPS capacity using data from the literature [20,27,40] (see Fig. 8). The calculated scale factor is 0.61 with an R^2 value of 0.9933 and indicates that capital cost increases at a slower rate than it should with an increase in capacity. Fig. 8 shows the capital cost for different capacities of MPS units. The capital cost is the cost for the biomass pretreatment unit, the pyrolysis unit, and the bio-oil condensing and bio-oil storage unit.

3.3. Biomass transportation distance in scenarios 1 and 2

In the two configurations (truncated and radial), the MPS unit is treated like a centralized unit for the specific times. In this study, four relocation times are considered: yearly, semi-annually, quarterly, and monthly. For the specific relocation time, biomass is transported to the MPS by 10 dry tonne capacity trucks. As stated earlier, the average transportation distance for the two scenarios depends on the MPS area. The MPS area varies with MPS capacity and relocation time. Fig. 9 (a) and (b) shows the biomass transportation distance for the truncated and radial scenarios, respectively. In both scenarios, biomass transportation distance increases with increases in capacity and relocation time. This distance is significantly higher in the radial scenario than in the truncated scenario. For a yearly relocation for a 100 dry t d^{-1}MPS capacity, transportation distance is 129 km for the radial scenario and 30 km for the truncated scenario. This is due to the nature of the assigned area. In the radial scenario, MPSs are located at the centroid of each sector and the centroid is located two-thirds the distance of the radius. But in the truncated scenario, the MPS is located in the center of the truncated d-segment, thus the transportation distance is considerably smaller than in the radial scenario.

3.4. Bio-oil transportation distance in scenarios 1 and 2

After the pyrolysis operation, bio-oil is transported to the upgrading facility in the center of the harvest area. The transportation distance varies with the MPS allocation configuration, as explained earlier. Fig. 10 (a) and (b) shows the average bio-oil transportation distance for both scenarios. In both cases, with changes in relocation time, a small change in bio-oil transportation distance is observed, in the range of 0.5–1 km. With variations in relocation frequency from yearly to monthly for a 100 dry t d^{-1} MPS capacity, the transportation distance changes from 10.80 to 10.90 km for both scenarios, which is insignificant. However, the change is somewhat significant at lower capacities. This is because the number of units increases the radius of the circle a little due to the addition of the settle-down area for each MPS. At lower capacities, the number of MPS units is comparatively higher than at higher capacities, and thus the bio-oil transportation distance increases slightly with the additional settle down areas.

Table 3
Field cost, transportation cost, and storage cost for an MPS unit.

Parameters	Value	Reference/Comments
Field cost		
Harvesting cost (cutting, chipping, and skidding) ($ t^{-1})	24.42	[34,35]
Premium paid to the land owner ($ t^{-1})	5.26	[18,35]
Road constuction and infrastucture cost ($ ha^{-1})	3.68	[18,35]
Silviculture cost ($ ha^{-1})	119.04	[18,35]
Transportation cost]		
Biochar transportation cost ($ t^{-1} of biochar)	4.5	[37]
Biomass transportation cost ($ m^{-3})	2.35 + 0.03*D*	*D* is the round-trip distance travelled by the truck from the forest to the plant gate [18,23,24,35].
Truck capacity for the ransportation of biomass dry tonne	10	[31,39]
Bio-oil transportation cost ($ m^{-3})	6.824 + 0.0.060*D*	*D* is the round-trip distance travelled by the liquid tank truck from the MPS point to the bio-oil processing plant [38].
Capacity of liquid tank truck (m^3)	30	[38]
Storage system		
Biomass storage type	On-field storage	[35]
Biomass stoarage cost ($ t^{-1})	1.1	[35]
No. of biomass storage days	15	Assumed
Bio-oil storage capacity (m^3)	100	This study
Cost of storage tank for bio-oil ($)	137,689.02	[24,38]
Material type for bio-oil storage tank	Stainless steel	[38]
No. of days bio-oil can be stored	3	[38]

Table 4
MPS plant characteristics and cost input.

Parameters	Value	Reference/Comments
MPS characteristics		
MPS capacity (dry t d^{-1})	10,20, 40, 50 &100	Assumed for this study
Lifetime of plant (years)	10	
Running time of MPS (days)	273	
Use rate (%)	90	
Finalcial analysis method	Discounted cash flow	
Internal rate of return (%)	10	
No. of shifts per day	3	[20]
No. of employees per shift	2 or 3	[22]
Adminstarative staff	1	Assumed
Employee salary including benefits ($ h^{-1})	30	[18,20,28]
Scale factor for MPS capacity from 10 to 100 dry t d^{-1}	0.61	Calculated. The details are included in section 3.2.
Maintanance costs	3% of initial capital investment	[18]
Relocation frequency	Yearly, semi-annually , quarterly, and monthly	Assumed.
Fast pyrolysis results (wt%)		
Bio-oil	57%	[20]
Biochar	27%	
Noncondensable gases	16%	
Energy used in MPS		
Bio-oil used to produce electricity (wt%)	11% of the bio-oil yield	This reduces the bio-oil yield to 50% from 57% by weight.
Thermal energy		
Propane	25%	[20]
Noncondensable gases	75%	
Propane cost ($ L^{-1})	0.456	[43]

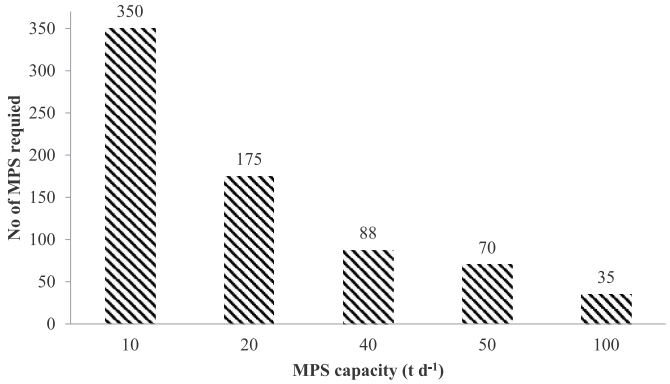


Fig. 7. Change in the number of MPS units with plant capacity.

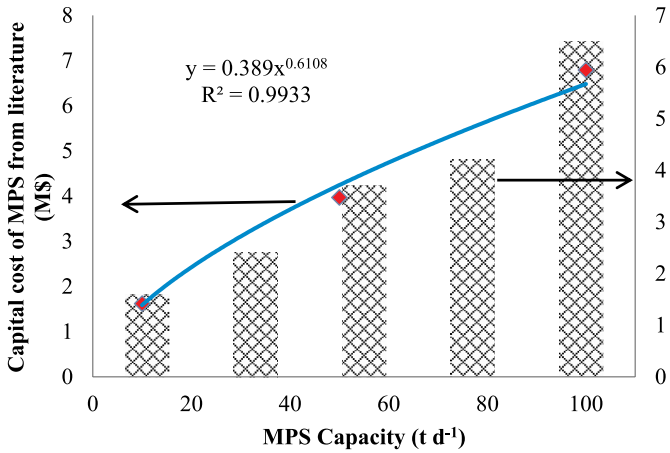


Fig. 8. The development of the scale factor for an MPS plant and changes in capital cost with MPS capacity.

3.5. Relocation distance

The relocation distances for scenarios 1 and 2 for different MPS capacities with different relocation times are shown in Fig. 11 (a) and (b). In both scenarios, the relocation distance is less than 2 km. The relocation distance is highest for the 100 dry t d^{-1} capacity yearly relocation for both scenarios. Because many MPSs work concurrently to achieve the base case target of 2000 dry t d^{-1} in a circular area, the relocation distance is significantly lower at that capacity for both scenarios. Therefore the relocation cost is considered an operating cost. To disassemble, relocate, and reassemble every MPS unit, 17 h of labor are required. Hence with an increase in the number of MPS units and relocation frequency comes an increase in relocation cost.

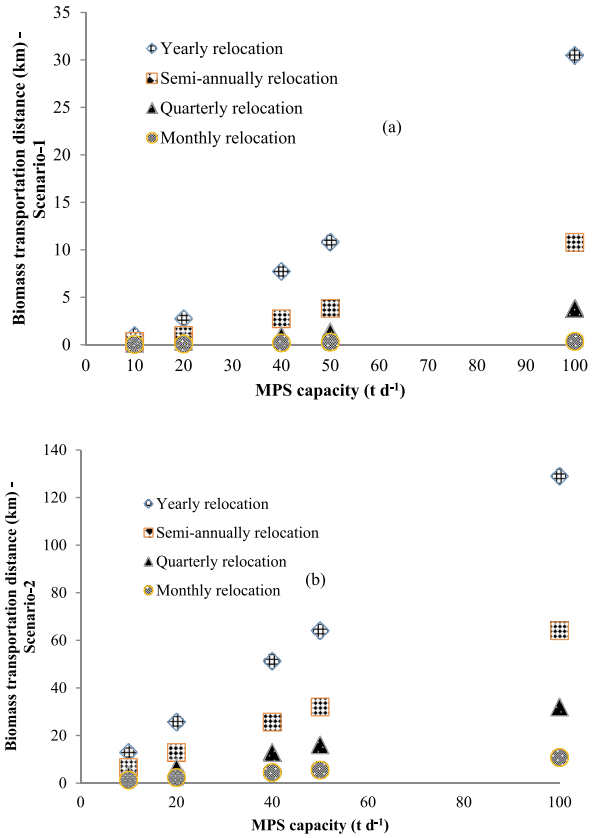


Fig. 9. Biomass transportation distance with changes in MPS capacity: (a) truncated and (b) radial configurations.

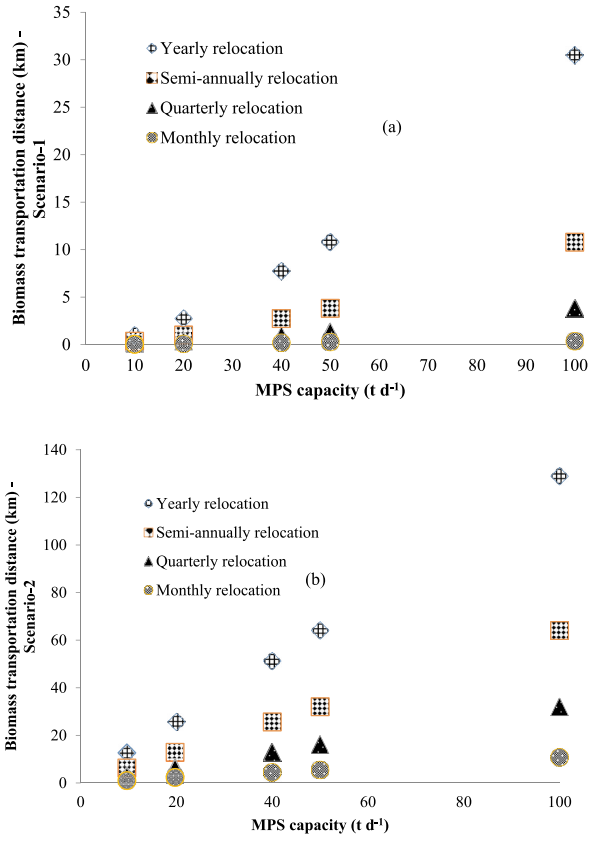


Fig. 10. Bio-oil transportation distance with changes in MPS capacity: (a) truncated and (b) radial configuration.

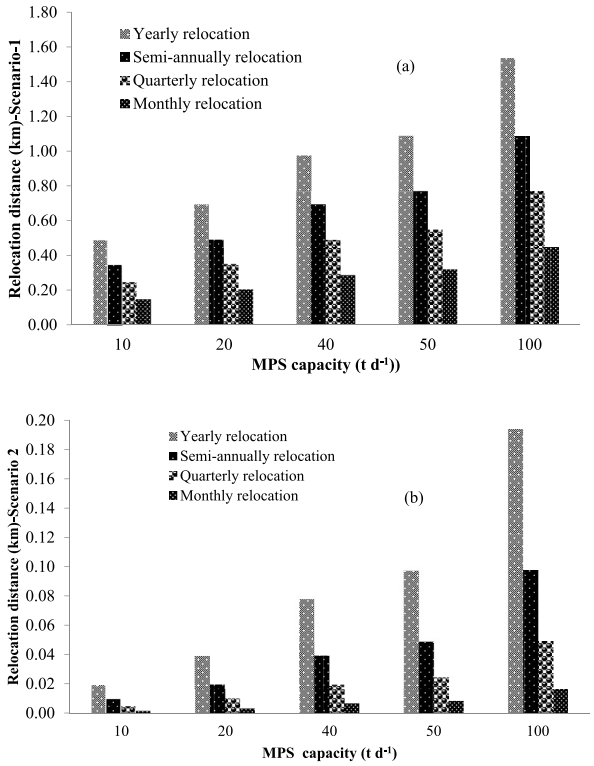


Fig. 11. MPS relocation distance with variations in MPS capacity: (a) truncated and (b) radial configurations.

3.6. Bio-oil production cost

This study investigates the bio-oil production costs of a 2000 dry t d^{-1} biomass plant for a mobile pyrolysis system and compares them with results for a centralized plant from work by Sarkar and Kumar [28]. The results are summarized in Table 5. Bio-oil produced in an MPS costs significantly more than bio-oil produced in a centralized plant because of economies of scale.

Of the two MPS scenarios, the truncated scenario shows better results than the radial scenario. Bio-oil production costs are drastically higher for low-capacity MPS units and decrease with increases in plant capacity for both MPS scenarios. The reason behind this is the labor cost. Even if the MPS capacity size changes, two employees are required to run the unit in each shift. Therefore, for lower capacity mobile units, far more units are required than for a higher capacity MPS unit. In a centralized unit, however, labor cost is not a significant component because the facility is fixed.

Table 6 shows the distribution of bio-oil production cost components for the radial and truncated scenarios for yearly relocations of a 2000 dry t d^{-1} plant. Of all the cost components, capital, labor, harvesting, and biomass transportation cost have a significant effect on the bio-oil production cost. With an increase in MPS capacity, labor costs fall from 76% to 46% and 75%–39%, respectively, for the truncated and radial scenarios. For a 10 dry t d^{-1}MPS unit, 350 units are active at one time. Each unit is operated by 2 employees per shift, which increases the labor cost significantly, resulting in very high labor costs compared to other cost components. Harvesting and biomass transportation costs increase with the increase in MPS capacity. For the harvesting cost, as the MPS capacity increases, it can handle more biomass than a lower capacity MPS. Therefore it requires more biomass as feed, which increases the harvesting cost. In the case of biomass transportation costs, a 100 dry t d^{-1} MPS unit has a greater transportation distance than the 10 dry t d^{-1} unit. But the biomass transportation distance is 12% for the truncated scenario and 23% for the radial scenario. The overall bio-oil production costs are lower for a 100 dry t d^{-1} plant with a yearly

Table 5
Bio-oil production costs in two frameworks, centralized and MPS.

Bio-oil production cost ($ L^{-1}) for 2000 t d^{-1} plant									
Capacity of MPS (t d^{-1})	Yearly relocation		Semi-annually relocation		Quarterly relocation		Monthly relocation		Centralized plant [28]
	Truncated	Radial	Truncated	Radial	Truncated	Radial	Truncated	Radial	
10	1.455	1.462	1.459	1.461	1.466	1.466	1.496	1.500	0.241
20	0.828	0.841	0.830	0.838	0.834	0.841	0.854	0.864	
40	0.502	0.528	0.503	0.523	0.506	0.526	0.520	0.543	
50	0.432	0.467	0.436	0.461	0.438	0.464	0.451	0.480	
100	**0.349**	**0.407**	**0.345**	**0.398**	**0.345**	**0.400**	**0.353**	**0.414**	

relocation: 0.294 $ L^{-1} and 0.334 $ L^{-1} for the truncated and radial scenarios.

Bio-oil production costs in the decentralized scenarios for a 100 dry t d^{-1} capacity and four relocation times are shown in Fig. 12. The bio-oil production cost decreases when the relocation time changes from yearly to semi-annually and with further reductions in location time from semi-annually to quarterly and monthly, bio-oil production increases. The cost distributions of different components are given in Table A1 in Appendix A. (Table A1)

3.7. Bio-oil cost variations with changing centralized plant capacity

Fig. 13 shows the variations in bio-oil production cost in the mobile pyrolysis system (100 dry t d^{-1}for yearly relocation) and the centralized system. It is interesting to observe that at lower capacities (less than 500 dry t d^{-1}), both MPS scenarios perform better than the centralized facility. With increases in plant capacity in the centralized plant, bio-oil production costs decrease due to economies of scale. But for the mobile system, the results are quite different. In the radial configuration, bio-oil production costs increase with an increase in base case plant capacity, and in the truncated configuration, production costs are almost constant with variations in plant capacity. The plot of the centralized plant intersects the radial and truncated scenarios at 410 and 500 dry t d^{-1}, respectively.

From this assessment it is reasonable to conclude that at lower capacities, it is good to adopt the MPS to produce bio-oil. In general, plant capacity in a region depends on the availability of biomass and the accessibility of the trees. If biomass resources are limited, it is better to use a mobile pyrolysis plant. And of the two MPS configurations considered, the truncated scenario is economically more attractive.

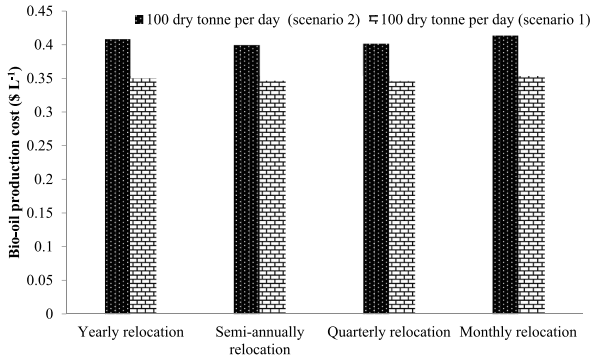


Fig. 12. Bio-oil production costs for a 100 dry t d^{-1} MPS plant in the truncated and radial configurations.

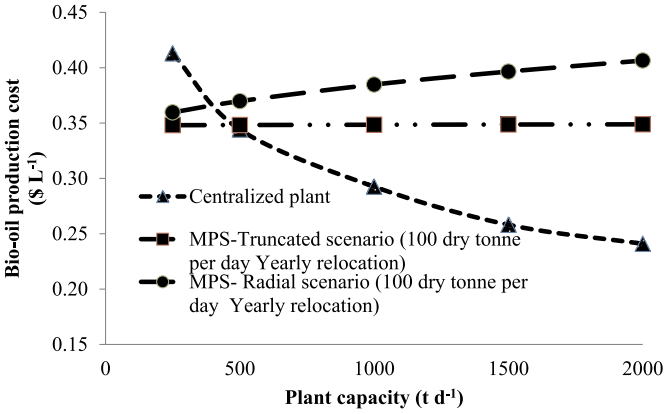


Fig. 13. Bio-oil production costs for centralized and decentralized pyrolysis plants.

Table 6
Cost distribution of bio-oil production costs for an MPS.

Cost distribution of bio-oil production costs ($ L^{-1}) for a 2000 dry t d^{-1} plant						
Cost parameters	Yearly relocation- Scenario 1:truncated			Yearly relocation- Scenario 2:radial		
	10 t d^{-1}	50 t d^{-1}	100 t d^{-1}	10 t d^{-1}	50 t d^{-1}	100 t d^{-1}
Capital cost	10.19%	18.73%	17.79%	10.14%	17.39%	15.12%
Labor and overhead cost	75.31%	51.87%	46.10%	74.94%	48.17%	39.20%
Harvesting cost	3.22%	11.10%	13.81%	3.21%	10.31%	11.74%
Transportation cost of biomass to MPS	1.68%	7.18%	12.46%	2.16%	13.79%	25.57%
Maintenance cost	2.04%	3.75%	3.56%	2.03%	3.48%	3.02%
Bio-oil transportation cost	0.58%	1.99%	2.48%	0.58%	1.85%	2.11%
Propane cost	5.73%	3.94%	2.45%	5.70%	3.66%	2.09%
Relocation cost	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Bio-oil storage cost	0.17%	0.12%	0.07%	0.17%	0.11%	0.06%
Biomass storage cost	0.88%	0.61%	0.38%	0.88%	0.56%	0.32%
Biochar transportation cost	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Bio-oil production cost ($ L^{-1})	**1.45**	**0.43**	**0.35**	**1.46**	**0.47**	**0.41**

3.8. Uncertainty analysis

A Monte Carlo analysis was performed on the 100 dry t d^{-1}MPS unit for a yearly relocation for the two configurations where bio-oil production cost was optimum. For these analyses, 10,000 iterations were considered and a triangular distribution was used. The input parameters considered were capital cost, labor and overhead cost, harvesting cost, transportation cost of biomass to the MPS, maintenance cost, bio-oil transportation cost, propane cost, relocation cost, bio-oil storage cost, biomass storage cost, and biochar transportation cost. The

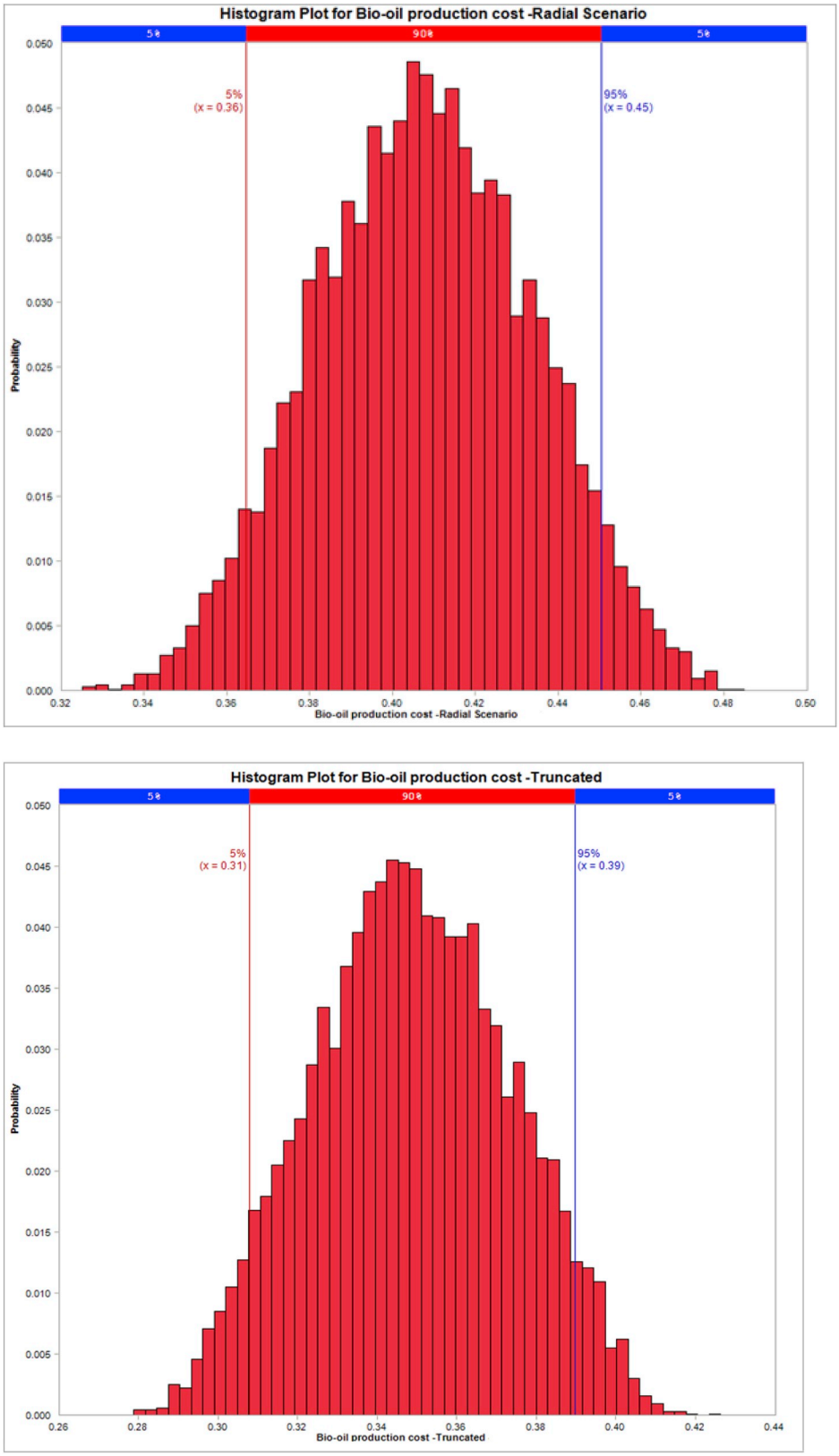


Fig. 14. Uncertainty analysis for bio-oil production cost for a 100 dry t d^{-1} capacity.

uncertainties considered for these variables were between 70 and 130%. The production cost of bio-oil was found to be 0.41 ± 0.0198 $ L^{-1} and 0.35 ± 0.0189 $ L^{-1} at a 95% confidence level, respectively, for radial and truncated configurations of a 100 dry t d^{-1} MPS unit for yearly relocation and is shown in Fig. 14. The uncertainty associated with the radial configuration was little higher than the truncated, which can be attributed to the higher standard deviation associated with the cost at 95% confidence.

4. Conclusion

Bio-oil can be produced from both centralized and decentralized pyrolysis plants and be further processed to produce transportation fuel. In this study, two supply chain networks were explored, radial and square, in which a number of decentralized units work simultaneously. From the cost distribution it was determined that by reducing the MPS capacity from 100 to 10 dry t d^{-1}, labor costs would increase from around 40%–75%. This is because with the decrease in capacity, the number of MPS units required increases, which increases labor costs. Of the two MPS scenarios, the truncated scenario is more economical than the radial scenario because it has a shorter biomass transportation distance.

Bio-oil production costs of the centralized and mobile frameworks were compared. For higher plant capacities, the centralized plant performs better because of economies of scale. For a 2000 dry t d^{-1}biomass plant, bio-oil production costs are 0.24 $ L^{-1}, 0.35 $ L^{-1}, and 0.41 $ L^{-1} for a centralized plant, an MPS unit in the truncated scenario (100 dry t d^{-1} unit, yearly relocation), and an MPS unit in the radial scenario (100 dry t d^{-1} unit, yearly relocation), respectively. For plant capacities below or equal to 500 dry t d^{-1}, an MPS performs better than a centralized plant. At low capacities, the capital and administrative costs in a centralized plant are significantly higher than those for an MPS. Therefore, at lower base case plant capacities, an MPS is recommended as it is movable, provides access to remote locations, and tolerates extreme weather conditions.

The MPS configurations developed to estimate bio-oil production costs in this study are generic and can be applied to any real-time development scenario or jurisdiction through location-specific modifications. Biomass cost, labor cost, truck capacity, energy cost, and capital cost will change with location. In general, biomass cost is a combination of biomass harvesting cost, biomass collection cost, and biomass transportation cost. Harvesting technology varies by feedstock and makes a significant contribution to bio-oil cost. In this study, a generic equation was developed for the biomass transportation cost based on the configuration of the selected area. To make the estimation of biomass transportation distance real, a geographical information system (GIS) can be used to calculate the exact distance between the biomass location and the MPS. Finally, labor cost and inflation are completely driven by the location of the plant and the benefits provided by the employer. These are the cost components that need to be changed when this technology is adopted in other jurisdictions. The concept and application of technology will be same irrespective of the location.

A centralized plant performs better at higher capacities, but it might not be preferred given the limited availability of large amounts of biomass feedstock. Biomass availability is the driver for the selection of the pyrolysis framework. The outcomes from this paper can also be used to encourage local governments to provide decentralized units to farmers, municipalities, and forest communities to divert waste and woody biomass to valuable products and generate employment at a small-scale pyrolysis unit.

Acknowledgments

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Appendix A

Scenario 1: Truncated

Calculation of angle φ:

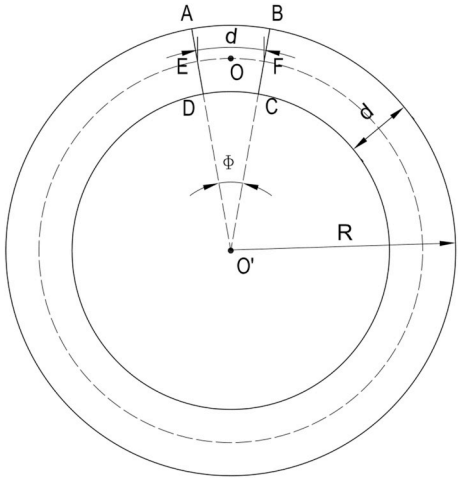


Fig. A1. Truncated scenario.

The perimeter of the dotted circle is given by Eqn. (14):

$$P = 2\pi^{*}\left(R - \frac{d}{2}\right) \tag{14}$$

The perimeter P creates an angle 2π at the center. So, angle φ (radian) is estimated by Eqn. (15):

$$\varphi = 2\pi^{*}\left(\frac{d}{P}\right) = 2\pi^{*}\frac{d}{2\pi\left(R - \frac{d}{2}\right)} = \frac{d}{\left(R - \frac{d}{2}\right)} = \frac{d}{\left(R_0 + \frac{d}{2}\right)} \tag{15}$$

where R = $\left(R - \frac{d}{2}\right)$ = $\left(R_0 + \frac{d}{2}\right)$ as shown in Fig. A3.

The area of the ring created by R and R-d is A_d as represented by Eqn. (16).

$$A_d = \pi R^2 - \pi(R - d)^2 = 2\pi Rd - \pi d^2 \tag{16}$$

The area of segment ABFCDEADEA is A_s (Fig. A1) and is given in Eqn. (17)

$$A_s = \frac{A_d \varphi}{2\pi} = \frac{2\pi Rd - \pi d^2}{2\pi} * \frac{d}{\left(R - \frac{d}{2}\right)} = d^2 \tag{17}$$

Bio-oil transportation distance

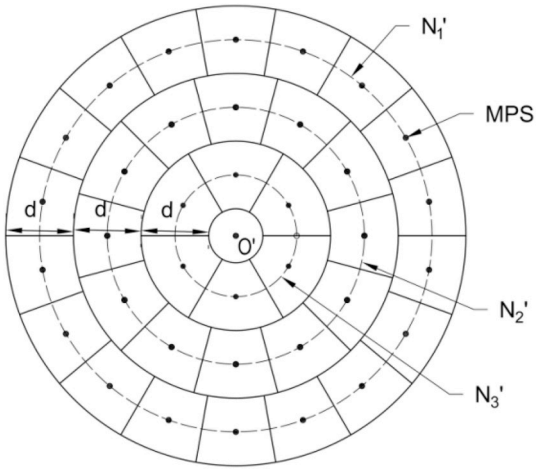


Fig. A2. Bio-oil transportation distance in the truncated scenario.

The number of all truncated segments in the 1st ring can be calculated using Eqn. (18).

$$N_1' = \frac{Perimeter\ of\ dotted\ circle\ N_1'}{d} = \frac{2\pi\left(R - \frac{d}{2}\right)}{d} \tag{18}$$

The number of all truncated segments in the 2nd ring can be calculated by using Eqn. (19).

$$N_2' = \frac{Perimeter\ of\ dotted\ circle\ N_2'}{d} = \frac{2\pi\left(R - \frac{3d}{2}\right)}{d} \tag{19}$$

The bio-oil transportation distance for all truncated segments located in the 1st ring can be calculated by using Eqn. (20).

$$D_1 = N_1'^{*}\left(R - \frac{d}{2}\right) = \frac{2\pi\left(R - \frac{d}{2}\right)}{d} * \left(R - \frac{d}{2}\right) = \frac{2\pi\left(R - \frac{d}{2}\right)^2}{d} \tag{20}$$

The bio-oil transportation distance for the d squares located in the 2nd ring can be calculated by using Eqn. (21).

$$D_2 = N_2'^{*}\left(R - \frac{3d}{2}\right) = \frac{2\pi\left(R - \frac{3d}{2}\right)}{d} * \left(R - \frac{3d}{2}\right) = \frac{2\pi\left(R - \frac{3d}{2}\right)^2}{d} \tag{21}$$

Biomass transportation distance

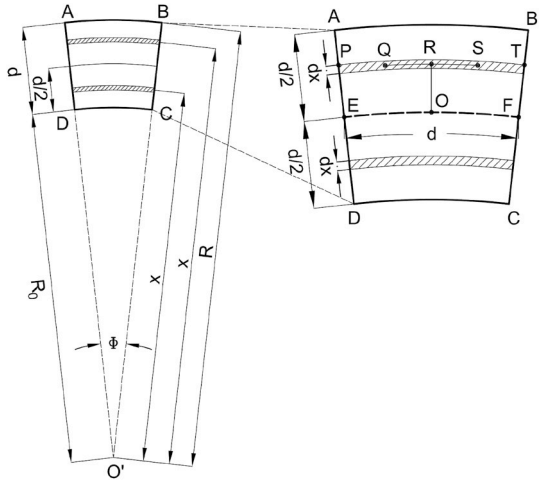


Fig. A3. Biomass transportation distance in the truncated scenario.

For the small area dx in the first segment, *ABFEA* (outer arc to MPS), the distance travelled by the truck to transport the biomass to the MPS unit is calculated by Eqn. (22):

$$\textit{Biomass transportation distance by truck} = d_1 = (QR + RO) + (SR + RO) = 2*(QR + RO) = 2*\left[\left(x*\frac{\varphi}{4}\right) + \left(x - R_0 - \frac{d}{2}\right)\right]$$
$$= 2*\left[\left(\frac{d}{\left(R_0 + \frac{d}{2}\right)}*\frac{x}{4}\right) + \left(x - R_0 - \frac{d}{2}\right)\right] \quad (22)$$

For the small area dx in the second segment, *EFCDE* (MPS to inside arc), the path followed by the truck to transport the biomass to the MPS unit is calculated by Eqn. (23):

$$\textit{Biomass transportation distance by truck} = d_2 = 2*\left[\left(\frac{d}{\left(R_0 + \frac{d}{2}\right)}*\frac{x}{4}\right) + \left(R_0 + \frac{d}{2} - x\right)\right] \quad (23)$$

The area of the small strip dx is given in Eqn. (24):

$$dA = (PR + RT)dx = 2*PR*dx = 2*\frac{\varphi}{2}*x*dx = \frac{d}{\left(R_0 + \frac{d}{2}\right)}*x*dx \quad (24)$$

The truck capacity for biomass transportation is 20 green tonnes per trip and a yield of 84 dry tonnes of forest biomass is assumed. The area required to collect biomass for one trip is estimated with Eqn. (25) and the number of trips required to deliver the biomass is given in Eqn. (26).

$$\textit{Area required for each truck trip} = \frac{\textit{Capacity of truck}}{\textit{Whole tree yield}} = \frac{10}{84}\frac{ha}{trip} \quad (25)$$

$$\textit{Number of truck trips required for the area} = \frac{dA}{10/84} = \frac{84\,dA}{10}\ trip/ha \quad (26)$$

The total distance travelled by truck and the number of trips for the area dA in segments 1 and 2 are given in Eqn. (27) and Eqn. (28), respectively.

$$dd_1 = \frac{84\,dA}{10}*2*\left[\left(\frac{d}{\left(R_0 + \frac{d}{2}\right)}*\frac{x}{4}\right) + \left(x - R_0 - \frac{d}{2}\right)\right] = 2*\frac{84}{10}*\left[\left(\frac{d}{\left(R_0 + \frac{d}{2}\right)}*\frac{x}{4}\right) + \left(x - R_0 - \frac{d}{2}\right)\right]*\frac{d}{\left(R_0 + \frac{d}{2}\right)}*x dx \quad (27)$$

$$dd_2 = 2*\frac{84}{10}*\left[\left(\frac{d}{\left(R_0 + \frac{d}{2}\right)}*\frac{x}{4}\right) + \left(R_0 + \frac{d}{2} - x\right)\right]*\frac{d}{\left(R_0 + \frac{d}{2}\right)}*x dx \quad (28)$$

To calculate the average biomass transportation distance for the whole area *ABFCDEA*, Eqn. (27) and Eqn. (28) are integrated from $R_0 + d/2$ to $R_0 + d$ and R_0 to $R_0 + d/2$, respectively. The total average distance travelled by truck can be estimated by summing the two integrations, which are estimated in Eqn. (29)

$$D=\int_{R_0+\frac{d}{2}}^{R_0+d} dd_1+\int_{R_0}^{R_0+\frac{d}{2}} dd_2=\int_{R_0+\frac{d}{2}}^{R_0+d} 2*\frac{84}{10}*\left[\left(\frac{d}{\left(R_0+\frac{d}{2}\right)}*\frac{x}{4}\right)+\left(x-R_0-\frac{d}{2}\right)\right]*\frac{d}{\left(R_0+\frac{d}{2}\right)}*x dx+\left[\int_{R_0}^{R_0+\frac{d}{2}} 2*\frac{84}{10}*\left[\left(\frac{d}{\left(R_0+\frac{d}{2}\right)}*\frac{x}{4}\right)+\left(R_0+\frac{d}{2}-x\right)\right]*\frac{d}{\left(R_0+\frac{d}{2}\right)}*x dx\right]$$

$$\Rightarrow D=\left[\frac{7d^3(6d^2+17dR_0+12R_0^2)}{5(d+2R_0)^2}\right]+\left[\frac{7d^5}{10(d+2R_0)^2}+\frac{7d^4}{10(d+2R_0)}+\frac{21d^4R_0}{5(d+2R_0)^2}+\frac{42d^3R_0^2}{5(d+2R_0)^2}+\frac{63d^3R_0}{5(d+2R_0)}+\frac{84d^2R_0^2}{5(d+2R_0)}-\frac{42d^2R_0}{5}\right]$$

$$\Rightarrow D=\frac{7d^3}{10(d+2R_0)^2}[13d^2+40dR_0+36R_0^2]+\frac{7d^3}{5(d+2R_0)}\left[\frac{d}{2}+3R_0\right] \quad (29)$$

Scenario 2: Radial scenario

Biomass transportation distance

For the small area d*r* in the first segment (from the center to the centroid), the distance travelled by the truck to transport the biomass to the MPS unit is calculated by Eqn. (30):

$$\text{Biomass transportation distance by truck } = d_1=(BN+NO)+(EN+NO)=2*(BN+NO)=2*\left[\left(r*\frac{\beta}{4}\right)+(r-R')\right] \quad (30)$$

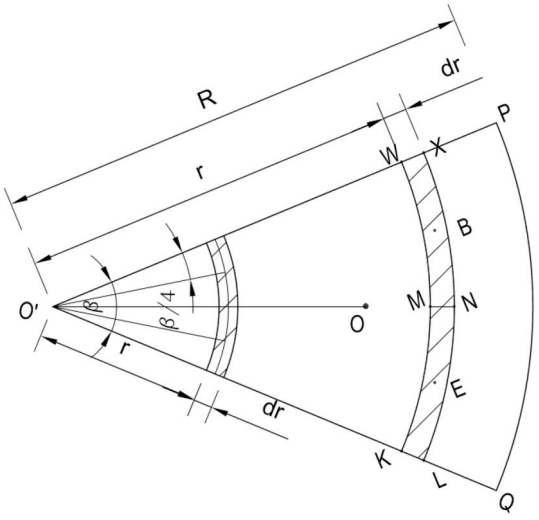


Fig. A4. Biomass transportation distance in the radial scenario.

For the small area d*r* in the second segment (from the centroid to the end of the arc), the path followed by the truck to transport the biomass to the MPS unit is calculated by Eqn. (31):

$$\text{Biomass transportation distance by truck } = d_2=2*\left[\left(r*\frac{\beta}{4}\right)+(R'-r)\right] \quad (31)$$

The area of the small strip d*r* is given in Eqn. (32):

$$dA=\frac{\pi(r+dr)^2}{Nm}-\frac{\pi r^2}{Nm}=\frac{2\pi r dr}{Nm} \quad (32)$$

As for the truncated configuration, the number of truck trips is estimated using Eqn. (26). The total distance travelled by truck and the number of trips for the area d*A* in segments 1 and 2 are given in Eqn. (33) and Eqn. (34), respectively.

$$dd_1=\frac{84\,dA}{10}*2*\left[\left(r*\frac{\beta}{4}\right)+(r-R')\right]=2*\frac{84}{20}*\frac{2\pi r dr}{Nm}*\left[\left(\frac{2\pi r}{4Nm}\right)+\left(r-\frac{2R}{3}\right)\right] \quad (33)$$

$$dd_2=\frac{84\,dA}{10}*2*\left[\left(r*\frac{\beta}{4}\right)+(R'-r)\right]=2*\frac{84}{20}*\frac{2\pi r dr}{Nm}*\left[\left(\frac{2\pi r}{4Nm}\right)+\left(\frac{2R}{3}-r\right)\right] \quad (34)$$

To calculate the average biomass transportation distance for the whole area *QO'P*, Eqn. (33) and Eqn. (34) are integrated from *R'* to *R* and *0* to *R'*, respectively. The total average distance travelled by truck can be estimated by summing the two integrations, which are estimated in Eqn. (35).

$$d=D=\int_{R'=\frac{2R}{3}}^{R} dd_1+\int_{0}^{R'=\frac{2R}{3}} dd_2=\int_{R'=\frac{2R}{3}}^{R} 2*\frac{84}{10}*\frac{2\pi r}{Nm}*\left[\left(\frac{2\pi r}{4Nm}\right)+\left(r-\frac{2R}{3}\right)\right]dr+\int_{0}^{R'=\frac{2R}{3}} 2*\frac{84}{10}*\frac{2\pi r}{Nm}*\left[\left(\frac{2\pi r}{4Nm}\right)+\left(\frac{2R}{3}-r\right)\right]dr$$

$$=\frac{2*84*2\pi}{10*Nm}*\frac{R^3}{27}\left[\left(\frac{9*2\pi}{4Nm}\right)+\frac{8}{3}\right] \quad (35)$$

Table A1
Cost distribution

	Scenario: Truncated 100 dry t d^{-1}			
	Monthly	Quarterly	Semi-annually	Yearly
Capital cost	16.7%	17.2%	17.1%	17.8%
Labour and overhead cost	46.0%	47.2%	47.1%	46.1%
Harvesting cost	13.8%	14.1%	14.1%	13.8%
Transportation cost of biomass to MPS	8.6%	10.0%	11.1%	12.5%
Maintenance cost	3.5%	3.6%	3.6%	3.6%
Bio-oil transportation cost	6.8%	3.8%	2.9%	2.5%
Propane cost	2.4%	2.5%	2.5%	2.5%
Relocation cost	0.0%	0.0%	0.0%	0.0%
Storage cost of bio-oil	0.9%	0.3%	0.1%	0.1%
Storage cost of biomass	0.4%	0.4%	0.4%	0.4%
Biochar transportation cost	0.0%	0.0%	0.0%	0.0%
Bio-oil production cost	0.9%	0.9%	0.9%	0.9%
	Scenario 2: Radial 100 dry t d^{-1}			
	Monthly	**Quarterly**	**Semi-annually**	**Yearly**
Capital cost	13.7%	15.4%	14.7%	15.1%
Labour and overhead cost	49.9%	39.8%	40.5%	39.2%
Harvesting cost	9.9%	11.1%	11.3%	11.7%
Transportation cost of biomass to MPS	11.1%	24.0%	24.5%	25.6%
Maintenance cost	3.6%	3.1%	3.1%	3.0%
Bio-oil transportation cost	5.3%	3.2%	2.5%	2.1%
Propane cost	3.8%	2.1%	2.2%	2.1%
Relocation cost	0.0%	0.0%	0.0%	0.0%
Storage cost of bio-oil	1.4%	0.3%	0.1%	0.1%
Storage cost of biomass	0.6%	0.3%	0.3%	0.3%
Biochar transportation cost	0.0%	0.0%	0.0%	0.0%
Bio-oil production cost	0.7%	0.8%	0.8%	0.8%

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